

Energizing the thermophysical properties of phase change material using carbon-based nano additives for sustainable thermal energy storage application in photovoltaic thermal systems

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ABSTRACT

As solar energy are intermittent in nature and not predictable, researchers and scientists are actively developing efficient thermal energy storage (TES) systems intending to maximize the utilization of solar energy. Phase change materials (PCM) are potential materials that are largely accessed towards TES. However, the notable drawback of PCM is their lower thermal conductivity, leading to slower heat transfer rates and reduced thermal energy storage density. Thus, the current study focuses on developing and exploring a PCM composite by embedding paraffin wax and graphene to enhance the heat transfer mechanisms, making it a promising option for TES applications. Various aspects of the composite's performance were examined, including its microstructural behaviour, chemical stability, thermal stability, thermal conductivity, thermal reliability, and heat transfer characteristics. The findings revealed that the inclusion of graphene led to a substantial increase of up to 75.09 % in thermal conductivity while preserving the melting enthalpy of the material. The newly developed nanocomposite also demonstrated chemically and thermally stable up to a temperature of 210 °C, and the thermal stability was slightly enhanced by adding nanoparticles. This nanocomposite also exhibited improved optical absorbance and reduced transmittance, enhancing its potential for solar energy absorption. It further demonstrated durability, maintaining stability even after undergoing 500 thermal cycles. Notably, the overall efficiency of the nano-enhanced PCM integrated photovoltaic-thermal system (PVT) enhanced by 29 % and 49 % greater than the PVT system and conventional PV system. Given these exceptional characteristics and performance enhancements, this nanocomposite material holds promise for significantly advancing future sustainable TES technologies.

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1. Introduction

The increasing use of fossil fuels and the resulting problem of global warming has spurred greater attention towards adopting renewable energy sources such as solar, wind, and ocean hydropower [1]. Solar energy stands out among these options due to its abundance, sustainability, freely accessible and environmental friendly [2]. Photovoltaic

Bahiraei et al. [19] demonstrated that graphite-dispersed PCM at 7.5 and 10 wt% achieved 620 % and 1100 % thermal conductivity enhancements through numerical investigations, respectively. From the above findings, the carbon-based nano-enhanced PCM is a potential material for PVT applications. Many researchers [20–24] reported that incorporating nanocomposite PCM extremely enhanced electrical and thermal performance.

Nomenclature		Nanocomposite codes	
<i>Abbreviations</i>		RT50	Paraffin wax
DSC	Differential scanning calorimetry	RT50-0.1Gr	Paraffin wax with 0.1 wt% graphene
EDX	Energy dispersive X-ray	RT50-0.3Gr	Paraffin wax with 0.3 wt% graphene
FTIR	Fourier transform infrared	RT50-0.5Gr	Paraffin wax with 0.5 wt% graphene
GNP	Graphene nanoplatelets	RT50-0.7Gr	Paraffin wax with 0.7 wt% graphene
Gr	Graphene	RT50-0.9Gr	Paraffin wax with 0.9 wt% graphene
MWCNT	Multiwalled carbon nanotubes	NeOPCM	Nano-enhanced organic phase change materials
PCM	Phase change materials	<i>Symbols</i>	
PV	Photovoltaic	α	Absorbance
PVT	Photovoltaic thermal	λ	Wavelength (nm)
SDG	Sustainable development goals	η	Efficiency (%)
SEM	Scanning electron microscope	τ	Transmittance
TES	Thermal energy storage	ρ	Density (kg/m ³)
TGA	Thermogravimetric analyzer		
UV–Vis	Ultraviolet–visible		

thermal systems (PVT) are used to harvest electrical and thermal energy from solar radiation. Typically, the generated electrical energy is stored in batteries, while thermal energy is stored in materials known as thermal energy storage (TES) materials [3]. Phase change materials (PCMs) are becoming popular TES due to their capacity to store and release latent heat at constant temperatures, enabling higher energy storage density [4]. Although, PCMs have numerous potential applications because of their storage capacity and ability to deliver heat evenly. However, their limited thermal conductivity imposes constraints on the rate of heat absorption and release rate. This limitation affects the solidification and melting process, making it slower [5,6]. The existing literature highlights several strategies to enhance the thermal conductivity of PCMs, such as dispersing nanoparticles [7,8], incorporating metal fins [9,10], using metal foams [11,12] employing multiple PCMs [13,14], and integrating heat pipes [15]. Presently, the research in this area is heavily focused on dispersing carbon-based nanoparticles within PCMs.

Sun et al. [7] examined the thermal performance of coconut shell charcoal and graphite nanopowder combined with PCM at different concentrations. They discovered that incorporating 0.02 wt% accelerated the melting process and improved thermal conductivity. They observed the addition of oleic acid dispersant effectively prevented agglomeration. The nanocomposite containing 0.06 wt% nanocarbon and 2.0 wt% oleic acid exhibited reported highest latent heat storage rate. Similarly, Reji Kumar et al. found that adding functionalized carbon-based nanoparticles to salt hydrate PCM significantly enhanced thermal and optical performance. The addition of 0.7 wt% nanoparticles to PCM increased thermal conductivity by 100 % [16]. Recent work by Hayat et al. [17] compared the energy storage and heat transfer performance of three different nanoparticles multiwalled carbon nanotubes (MWCNTs), graphene nanoplatelets (GNP), and TiO₂ dispersed in paraffin PCM at varying concentrations. The most significant enhancement in thermal conductivity (170 %) was achieved using 1.0 wt% GNP + MWCNT-PCM. Du et al. [18] performed molecular simulations to investigate the thermal performance of carbon-based nanoparticle-enhanced PCM for potential TES applications, revealing a significant increase in thermal conductivity compared to pure PCM.

Recent studies have also incorporated carbon-based nanoparticle-dispersed PCM into PVT systems, yielding enhanced energy performance [25]. Incorporating 5 wt% of GNP-PCM led to a 3.9 % reduction in photovoltaic panel temperature and a 9.2 % increase in electrical efficiency. Further research by Islam et al. [26] examined nano-enhanced organic PCM (NeOPCM)-integrated PCM within PVT systems, demonstrating significant improvements in panel temperature, energy, and exergy performance. Al-Waeli et al. [27] analyzed the energy performance of NeOPCM integrated PVT system with nanofluid to improve the energy performance. It was noted that a significant reduction in panel temperature, correspondingly, substantial improvement in thermal and electrical efficiency by 72.0 % and 13.7 %, respectively. Jamil et al. [20] examined the performance of PVT systems integrating with silica and carbon black dispersed PCM. It has been noted that the maximum temperature reduction of the PVT system with SiO₂/PCM and carbon black/PCM by 8.82 °C and 9.74 °C, respectively, improved electrical efficiency by 12 % and 12.07 %, respectively. Liu et al. [21] explored indoor PVT systems with nano-enhanced PCM, revealing reduced panel temperatures and overall system efficiency improvements. In addition, In addition, MWCNT dispersed PCM and copper foam matrix is used to regulate the PV system's temperature and enhance the efficiency. The integration of NePCM improves the electrical output from 4 % to 21 % due to the reduction of PV panel temperature [28].

Despite extensive research on nanocomposite PCM, graphene nanoparticles stand out due to their superior thermal conductivity and surface area. In this research, the attempt to improve the energy storage/release rate of RT50 PCM (melting point 50 °C) by incorporating graphene nanoparticle. To ensure the thermophysical properties of graphene-dispersed RT50 PCM, we conduct thermal stability, chemical stability, light absorptance, bandwidth, melting enthalpy, thermal conductivity and reliability analysis. The experimental thermal conductivity was compared with the numerical analysis values to compare with base PCM and developed nanocomposite PCM. To ensure the reliability of the nanocomposite, a 500-heating cooling cycle has tested and examined the thermal and chemical performance. In addition, a numerical heat transfer analysis was investigated with various input heat

sources to ensure the developed nanocomposite's heat transfer rate. The optimized nanocomposite is also integrated with PVT systems to analyze the electrical and thermal performance. The enhanced thermal conductivity, latent heat, light absorption and thermal stability of developed nanocomposite make it a promising material for thermal management in battery, PV systems and waste heat recovery applications [29–31]. The findings of this research are anticipated to contribute towards United Nations Sustainable Development Goals 7 and 13, aiming for affordable and clean energy and climate action.

2. Materials and methods

2.1. Materials used

Paraffin wax RT50, with a melting temperature of 50 °C, was acquired from Rubitherm, Germany. The melting enthalpy value of RT50 is stated to be 160 J/g and without any supercooling issues. Graphene (Gr) nanopowder with an average particle size of 8 nm was sourced from a graphene supermarket. Table 1 summarises the thermophysical properties of RT50 and graphene nanopowder.

2.2. Sample preparation

A two-step methodology was employed to formulate the Gr-infused organic PCM as shown in Fig. 1. The requisite RT50 PCM and Gr nanopowder (0.1, 0.3, 0.5, 0.7, and 0.9 wt%) amount was accurately measured using a microbalance (EX224, OHAUS). The measured PCM was placed in a beaker and heated on a hot plate (RCT, BASIC IKA) until it exceeded its melting temperature. Subsequently, the measured Gr nanopowder was introduced into the liquid PCM. The liquid mixture undergoes sonication in a bath sonicator (Ultrasonic Bath EASY 60H, ELMASONIC) for 90 min to ensure uniform dispersion. Throughout this process, the hot plate's temperature was maintained above the melting point of the PCM. The sample was allowed to cool down to room temperature after sonication naturally. The procedure was replicated to produce nanocomposites with varying Gr concentrations (0.1, 0.3, 0.5, 0.7, and 0.9 wt%). The distinct nanocomposites were designated as follows: paraffin wax (RT50), paraffin wax with 0.1 wt% Gr (RT50-0.1Gr), paraffin wax with 0.3 wt% Gr (RT50-0.3Gr), paraffin wax with 0.5 wt% Gr (RT50-0.5Gr), paraffin wax with 0.7 wt% Gr (RT50-0.7Gr), and paraffin wax with 0.9 wt% Gr (RT50-0.9Gr).

2.3. Characterization and instruments used

An array of instruments was employed to characterize the formulated nanocomposites' physical, chemical, and thermal properties. Morphological and microstructural characteristics of Gr, PCM, and NeOPCM were analyzed using a scanning electron microscope (SEM) (Model: TESCAN VEGA-3). Energy dispersive X-ray spectroscopy (EDX) (Model: Oxford Instruments) integrated with the SEM was used to mapping of element distribution within the composite. Chemical stability was analyzed via Fourier transform infrared (FTIR) spectroscopy (Model: PerkinElmer, Spectrum Two) spanning the wave number range of 400 cm⁻¹ to 4000 cm⁻¹. The degradation behaviour of the samples was studied using a thermogravimetric analyzer (TGA) (Model:

PerkinElmer, TGA 4000) at the temperature elevation from 30 °C to 400 °C. Optical performance was evaluated through ultraviolet–visible (UV–Vis) spectroscopy (Model: LAMDA 750, PerkinElmer) over a 200 nm–1400 nm wavelength span. The thermal conductivity of the samples was studied using a thermal properties analyzer (TEMPOS) (Meter Group, SH-3) at room temperature. Melting enthalpy was quantified via differential scanning calorimetry (DSC), operated at a temperature spectrum of 30 °C–70 °C under a nitrogen atmosphere, with a heating rate of 10 °C/min. The reliability of the samples was analyzed using a custom thermal cycler (Thermal cycler-ABBS). The cycles encompassed a temperature range of 30 °C–80 °C, with the heating element automatically deactivating upon reaching the maximum temperature and activating the cooling mechanism. Numerical heat transfer analysis of the developed nanocomposites was accomplished using ANSYS. The PLANE55 element possesses the flexibility to function as either a plane element or an axisymmetric ring element, offering a 2-D heat conduction capability. The element is characterised by four nodes, each possessing a single degree of freedom, which corresponds to the temperature variable. The element is suitable for conducting a two-dimensional investigation of thermal phenomena, whether in a steady-state or transient condition. The element is capable of compensate of mass transport heat flow originating from a uniform velocity field. With this option the thermal parameters are interpreted as analogous fluid flow parameters. The performance of the NeOPCM integrated PVT system was examined by indoor PVT simulator. The parameters of the photovoltaic system, encompassing short-circuit current, open-circuit voltage, and electrical power, were measured using a PV analyzer connected to a computer. Similarly, PV panel surface temperature, water inlet and outlet temperature were monitored utilizing K-type thermocouples connected to a data logger, with data acquisition occurring every 60 s. The range, accuracy, and uncertainty of the monitoring equipment are detailed in Table 2.

In this study, the uncertainty was measured by using Klein and McClintock method [32,33]. Equation (1) is used to calculate the uncertainty of the instruments used.

$$W_R = \left[\left(\frac{\partial R}{\partial x_1} w_1 \right)^2 + \left(\frac{\partial R}{\partial x_2} w_2 \right)^2 + \dots + \left(\frac{\partial R}{\partial x_n} w_n \right)^2 \right]^{0.5} \quad (1)$$

From the Klein and McClintock equation (1), the uncertainty of electrical efficiency measurement is.

$$WR_1 = [(0.06)^2 + (0.07)^2 + (1.0)^2] = 1.00.$$

The uncertainty of thermal efficiency measurement is.

$$WR_2 = [(1.0)^2 + (0.004)^2 + (0.14)^2 + (0.14)^2] = 1.04.$$

The uncertainty of overall efficiency measurement is

$$W_R = [(1.00)^2 + (1.04)^2]^{0.5} = 1.43$$

The uncertainty of the experimental test rig measuring instrument is lower than 2.0 %. The result confirmed that the measured devices are higher accuracy.

2.4. Numerical heat transfer analysis

The current study presents a comprehensive heat transfer analysis involving PCM and NeOPCM in response to supplied heat sources, as depicted in Fig. 2. To facilitate understanding, a two-dimensional enclosure was created and analyzed. The upper portion remained open, a constant heat source was connected to provide a steady heat input, and the remaining three sides were thermally insulated. Initially, heat source temperatures of 55 °C, 65 °C, 75 °C, and 85 °C were selected (all surpassing the melting temperature of the PCM). The entire system starts at room temperature, and subsequent heat transfer performance was investigated. We analyzed the heat transfer rate of PCM and NeOPCM with thermal conductivity 0.23 W/mK, and 0.403 W/mK.

Table 1
Thermophysical properties of RT50 and graphene.

Properties	RT50	Graphene
Melting enthalpy (J/g)	146	–
Melting temperature (°C)	50	3550
Thermal conductivity (W/mK)	0.2	4000
Density (kg/m ³)	880	2267
Specific heat capacity (J/gK)	2.0	2.1
Colour	Blueish white	Black

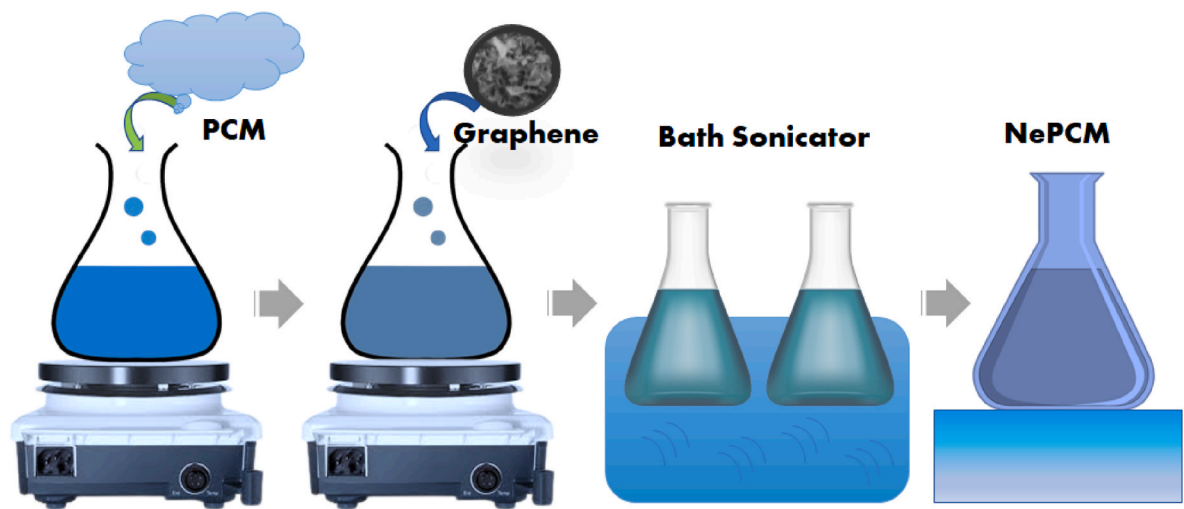


Fig. 1. Preparation of nano-enhanced organic phase change material.

Table 2
Accuracy and uncertainty of instruments.

Instrument	Range	Accuracy	Maximum uncertainty (in experiments)
Data logger	8 channels	±2 %	±1.0 °C
Flow meter (LPM)	0.2–2.0	±0.5 %	±4*10 ^{−3} LPM
Thermocouple	Pt-100 and K type	±0.5 %	±0.14 °C
Pyranometer (W/m ²)	0–2000 W/m ²	±5 %	±1.0 W/m ²
I–V tracer (V and A)	0–50 V and 0–16 A	±5 %	±0.06 V and ±0.07 A
Multimeter (V)	400 mV–1000 V	±0.4 %	±0.06 V
Water thermal sensor (°C)	1–1000	±0.5 %	±0.14 °C

2.5. Experimental and simulation setup

PVT systems are designed with the dual purpose of generating both electricity and thermal energy simultaneously. These systems consist of crucial components such as a photovoltaic panel and thermal collectors to generate electrical and thermal energy. Thus, the main objective of this study is to enhance the electrical performance of the PVT system while also capturing and storing thermal energy effectively. A solar simulator was utilized to evaluate the PVT system’s performance to achieve this goal. A 100 W polycrystalline silicon PV panel with 36 cells was commissioned in the study. The distinctive feature of this work is the incorporated copper tubes at the rear side of the panel in the PVT system. These tubes are firmly attached to the absorber plate using a thermal paste. The fully assembled PVT system was positioned within the solar simulator. Four distinct configurations- PV alone, PVT, PVT-PCM, and PVT-NeOPCM, were used, and their performances were assessed. While the performance of the PV and PVT systems was

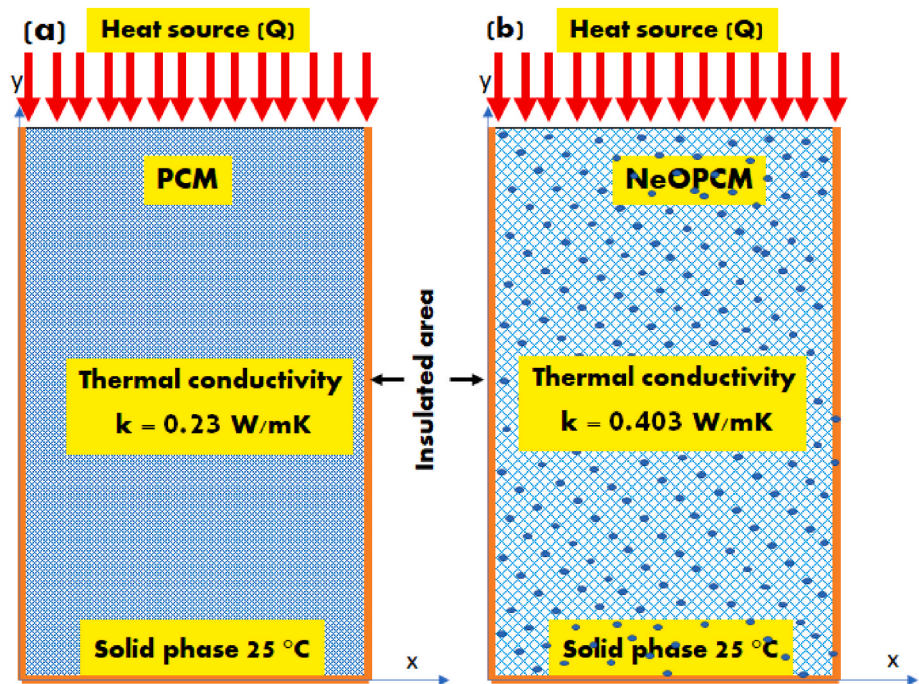


Fig. 2. Boundary condition of heat transfer rate (a) Pure PCM, and (b) RT50-0.7Gr nanocomposite.

experimentally evaluated using an indoor solar simulator, the PVT system integrated with PCM and NeOPCM was also analyzed through TRNSYS simulation. For the simulation, the solar radiation was set to 800 W/m^2 and a flow rate of 0.0125 kg/s .

Fig. 3 illustrates the schematic layout of the PVT system setup with the solar simulator. A fluid cycle connects the PVT system, allowing water (cooling fluid) to circulate within the tubes at the specified flow rate to release stored heat to an external heat exchanger. Improving the temperature regulation between the PV panel aids in improving the heat transfer process, thereby reducing the panel's temperature and enhancing the overall efficiency of the PVT system. Solar terminals are connected to the I-V tracer. A pyranometer was placed on top of the panel that linked to a data logger. All measurement instruments, including K-type thermocouples, flow meters, pyranometers, and I-V tracers, were integrated into the data logger and acquisition system, as depicted in **Fig. 3**. Multiple parameters were measured: ambient temperature, solar radiation, water inlet and outlet temperatures, PV panel top and rear side temperatures, flow rate, electrical power, short-circuit current, and open-circuit voltage. All instruments were calibrated against established standard or reference values before data collection.

The PVT system integrated with PCM and NeOPCM was also analyzed using the TRNSYS simulation software. TRNSYS is a widely recognized commercial software known for its ability to simulate the performance of both electrical and thermal systems. As there was no default library component available for PCM/NeOPCM-integrated PVT system, a new component named "Type 213" was developed using FORTRAN programming. Within the software framework, different simulation options are categorized as 'Types,' and the "Type 213" component was utilized in this research to model a glazed PV array integrated with PCM/NeOPCM for thermal energy storage applications.

In this simulation, the electrical performance is determined using a user-defined overall panel efficiency that can remain constant or vary based on solar radiation and cell temperature. The primary aim of this study was to enhance power output by regulating the PV surface temperature. Various parameters were input into the software for the simulation, as outlined in [Table 3](#). The results generated from the analysis contained power output, heat gain, thermal fluid outlet temperature, panel surface temperature, and PCM/NeOPCM temperature.

Table 3
Parameter used for TRNSYS simulation.

Parameters and input	Value
Area of PVT (m^2)	0.59
Irradiation (W/m^2)	800
Fluid flow rate (kg/s)	0.0125
Specific heat of fluid (J/kgK)	4190
Density of fluid (kg/m^3)	1000
Thermal conductivity of fluid (W/mK)	0.6
PCM thermal conductivity (W/mK)	0.23
NeOPCM thermal conductivity (W/mK)	0.403
Density of PCM (kg/m^3)	880
Specific heat of PCM (J/kgK)	2000
Latent heat of PCM (J/g)	136.00
Latent heat of NeOPCM (J/g)	137.93
Inlet water temperature ($^{\circ}\text{C}$)	25

2.6. Energy analysis

The energy analysis of the PVT-NePCM system under steady state is given by equation (3)

$$E_{in} = E_{out} + E_{loss} \quad (2)$$

$$E_s + E_{win} = E_{wout} + E_{el} + E_{loss} \quad (3)$$

where, E_{in} = Energy in, E_{out} = Energy out, E_s = The rate of solar irradiation interrupting the PV panel, E_{el} = Rate of electrical power production, E_{win} = Rate of heat flux water inlet, E_{wout} = Rate of heat flux water outlet, E_{loss} = Energy loss.

The PVT system thermal efficiency is calculated by equation (4) [26, 34].

$$\eta_{th} = \frac{Q_u}{A \times G} \quad (4)$$

$$Q_u = [mC_p(T_{out} - T_{in})] + [m_{NePCM}LH_{NePCM}] \quad (5)$$

where, η_{th} = thermal efficiency (%), Q_u = useful collected heat kJ/s, A = Area of PV unit in m^2 , G = Solar radiation W/m^2 , m = flow rate of cooling fluid (water) in kg/s, C_p = Specific heat of water in J/kg K, T_{out} and T_{in} = Outlet and inlet water temperature $^{\circ}C$, m_{NDPCM} = mass of NePCM in kg, LH = latent heat of NeOPCM J/kg.

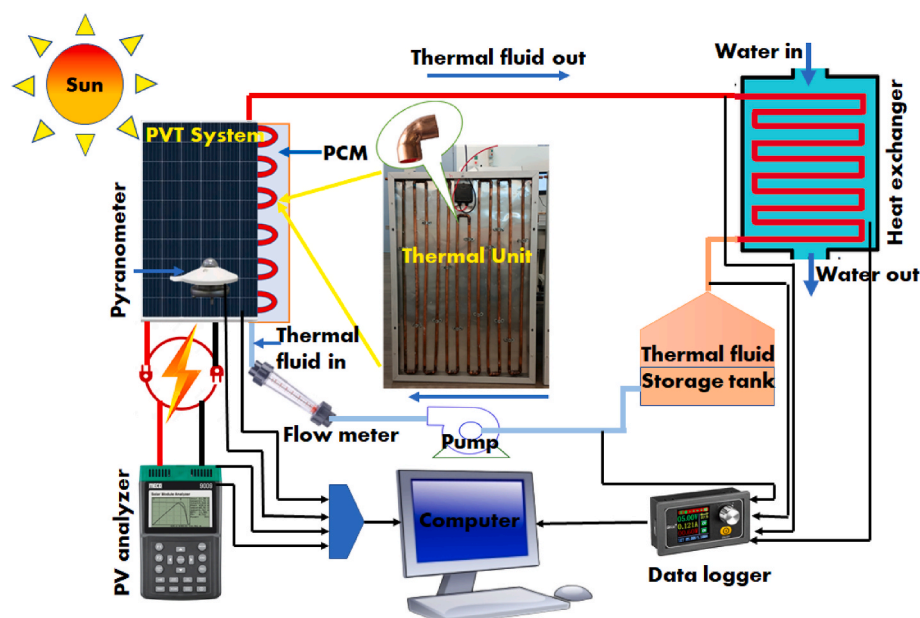


Fig. 3. Schematic diagram of the PVT system.

The required mass of NePCM can be calculated by equation (6) [26, 35].

$$m_{NDPCM} = \frac{Q_{hc}}{LH_{NDPCM} + \int_i^m C_{ps}(T)dT + \int_m^f C_{pl}(T)dT} \quad (6)$$

where, Q_{hc} = heat required for phase change (heat charging phase change) kJ/kg, LH_{NDPCM} = latent heat of NeOPCM in kJ/kg, C_{ps} & C_{pl} = Specific heat at solid & liquid PCM kJ/kg.K, i = initial temperature of NeOPCM, m = melting temperature of NeOPCM, f = final temperature of NeOPCM, dT = changing temperature.

The PVT system's electrical efficiency is calculated by equation (7) [26].

$$\eta_{el} = \frac{P_{max}}{A \times G} \quad (7)$$

$$P_{max} = I_{sc} \times V_{oc} \times FF = I_{mp} \times V_{mp} \quad (8)$$

where, η_{el} = electrical efficiency (%), P_{max} = maximum power (W), FF = Fill factor, I_{mp} = Current at maximum power (A), V_{mp} = Voltage at maximum power (V), The overall efficiency can be calculated by equation (9)

$$\eta_{over} = \eta_{th} + \eta_{el} \quad (9)$$

3. Results and discussion

The following section discusses various characterization results of Gr-dispersed PCM, thermal reliability, and numerical heat transfer rates. The optimal nanocomposite PCM was integrated with the PVT system, and its energy performance was thoroughly analyzed.

3.1. Microstructure and composition analysis

The morphological properties of PCM, nanoparticles, and nanocomposites were examined through acquired SEM images. Fig. 4 presents high-resolution images of PCM, Gr, and the resulting nanocomposite. Organic RT50 PCM is seen to possess a sticky nature owing to its alkane family attributes, as illustrated in Fig. 4(a). The microstructure of Gr nanoparticles, each measuring less than 10 nm, is displayed in Fig. 4(b). These nanoparticles exhibit a consistent flake-like structure, effectively occupying voids, pores, and the surfaces of the PCM. SEM images of the Gr-dispersed PCM composite are depicted in Fig. 4(c). It is clearly observable that the Gr nanoparticles are uniformly dispersed within a homogeneous mixture. Notably, some aggregation is observed among these nanoparticles within the alkane family context. The uniform dispersion of Gr nanoparticles suggests favourable properties for enhancing heat transfer through the formation of thermal

networks. An EDX analysis was performed alongside morphology analysis to verify the presence of Gr nanoparticles within the PCM composite. Fig. 5(a) shows the EDX spectrum of PCM, distinctly indicating the presence of carbon elements, as organic PCM primarily comprises carbon. The EDX spectrum analysis of the 0.7 wt% Gr-dispersed PCM composite is presented in Fig. 5(b). Evidently, both carbon and oxygen elements are detected within the composite. Moreover, the carbon content surpasses the PCM matrices, underscoring the successful integration of Gr nanoparticles.

3.2. Chemical stability

Investigating the wavelength location and intensity of absorption bands in the formulated carbon-based nanocomposite PCM is crucial in detecting chemical presence. The FTIR spectrum, depicted in Fig. 6, was pivotal in identifying the functional groups within the PCM and nanocomposites. For the pure PCM, four IR active vibrations were discerned at wavenumbers 718 cm^{-1} , 1464 cm^{-1} , 2844 cm^{-1} , and 2917 cm^{-1} , attributing these functional groups to the alkane group. These groups exhibited distinctive characteristics: a pronounced, sharp peak from $-\text{CH}_3$ symmetric stretching at wavenumber 2917 cm^{-1} , a strong $-\text{CH}_2$ symmetric stretching at 2844 cm^{-1} , a medium functional group encompassing $-\text{CH}_2$ and $-\text{CH}_3$ deformation bending at 1464 cm^{-1} , and a faint peak at 718 cm^{-1} signifying C-H rocking deformation bending. Likewise, the various weight percentages of Gr-dispersed PCM composites displayed analogous peaks and functional groups to those of the pure PCM. No new peaks or functional groups emerged. This observation implies that no chemical reactions transpired between the PCM and nanoparticles, thereby confirming the chemical stability of the developed nanocomposite [36]. This outcome aligns with findings reported by Kalidasan et al., who utilized organic PCM mixed with Ag nanoparticles [37].

3.3. Optical property

Employing PCMs is a highly effective method for storing thermal energy. PCMs actively store thermal energy by undergoing a remarkable phase transformation, subsequently releasing it in a controlled manner that aligns with their designated applications. Globally, solar energy is paramount due to its electromagnetic wave nature. PCMs are favoured for energy storage because they can replicate high irradiation absorption potential. Notably, organic PCMs are preferred due to their favourable properties in terms of transmittance, revealing exceptional radiance while possessing the unique ability to transmit light.

Solar radiation is typically categorized into three regions: violet (280–380 nm), visible (380–740 nm), and near infrared (740–1400 nm). The transmittance of PCM and various concentrations of Gr-dispersed

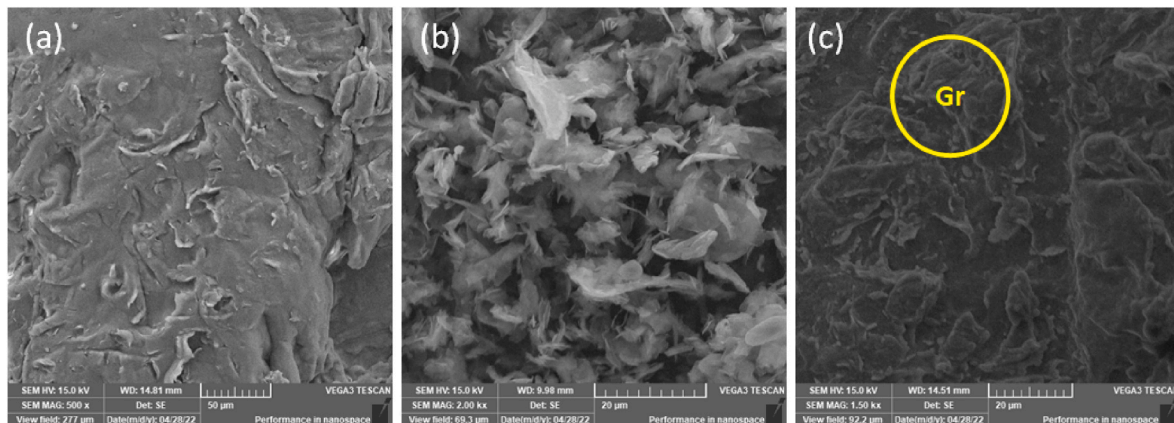


Fig. 4. SEM images of (a) Pure Paraffin wax RT50, (b) Gr powder, (c) Gr dispersed R50PCM composite.

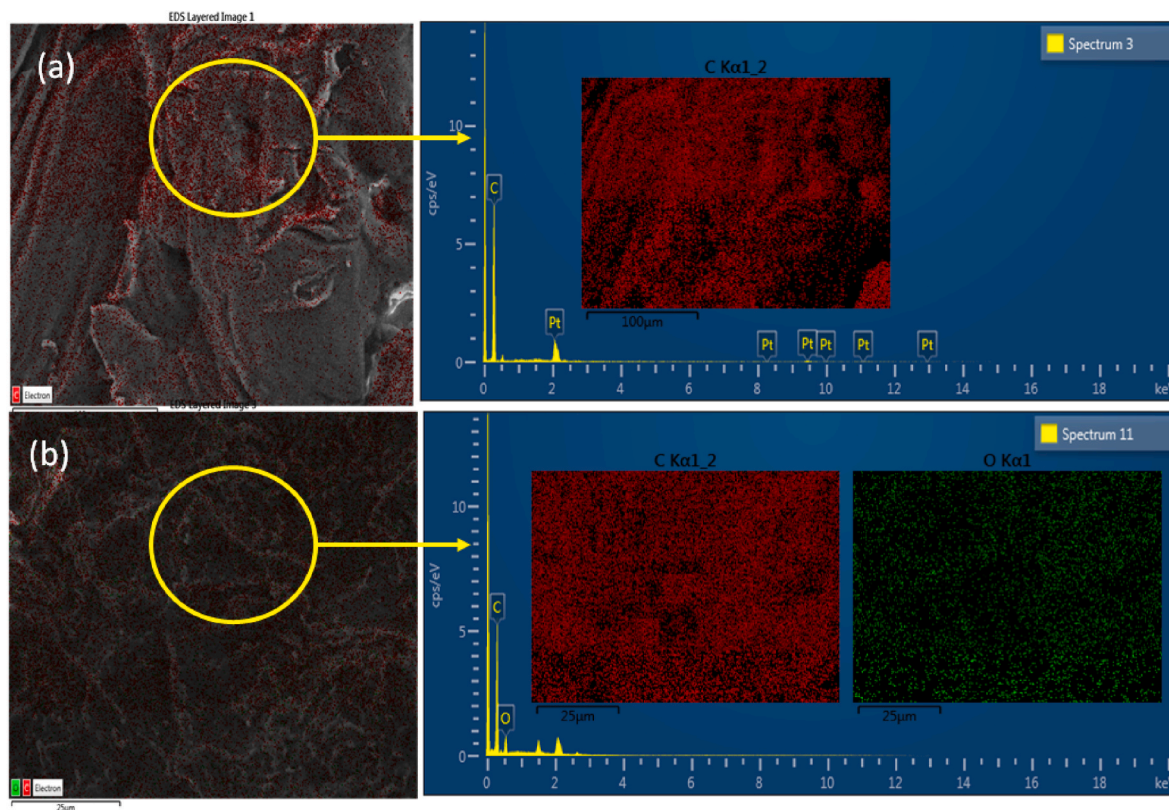


Fig. 5. EDX results of (a) Pure paraffin waxRT50, (b) Gr dispersed RT50 PCM composite.

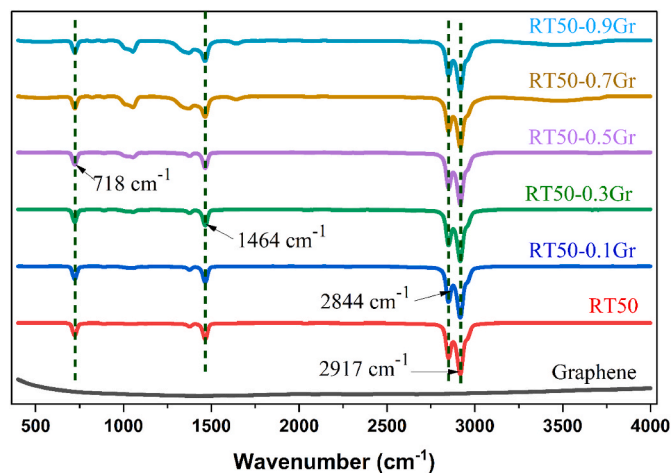


Fig. 6. FTIR results of RT50 PCM and RT50 with various concentrations of Gr.

nanocomposites is depicted in Fig. 7(a). However, introducing Gr into PCM decreases light transmission but enhances absorbance capacity. The transmittance percentage was calculated for wavelengths ranging from 250 to 1400 nm to analyze the light transmission characteristics of the developed composites in relation to solar spectrum data from NREI [38]. Pure PCM exhibited a transmittance of 22 %, while the developed nanocomposites (RT50-0.1Gr, RT50-0.3Gr, RT50-0.5Gr, RT50-0.7Gr, and RT50-0.9Gr) demonstrated transmittance percentages of 14.55 %, 14.34 %, 15.9 %, 15.39 %, and 15.145 %, respectively. This reduction in transmittance is attributed to incorporating carbon-based Gr nanoparticles into RT50 PCM.

Correspondingly, Fig. 7(b) illustrates the absorbance behaviour of PCM and the carbon nanoparticle dispersed PCM composite. The graph

clearly shows a substantial increase in absorbance capability due to the darker nature of the nanocomposites. The PCM exhibited an absorbance of around 0.63, whereas the nanocomposites (RT50-0.1Gr, RT50-0.3Gr, RT50-0.15Gr, RT50-0.17Gr, and RT50-0.19Gr) demonstrated absorbance values of around 0.82, 0.78, 0.80, 0.81, and 0.8, respectively. The percentage improvement in absorbance was calculated to be approximately 28.57 % for the 0.7 wt% Gr-dispersed PCM. The increased absorption observed in the nanocomposites can be attributed to their outstanding photothermal capability derived from their darker nature. When radiation interacts with a medium, it either gets absorbed, transmitted, or reflected. Organic PCMs, by nature, exhibit transparency that enables radiation to pass through. It is also clear that transmittance and absorbance share an inverse relationship. Thus, incorporating Gr into PCM leads to decreased transmission while enhancing absorbance. A higher absorptivity characteristic results in a more efficient response to incident irradiation, expediting thermal energy storage.

3.4. Bandgap

The absorption peak observed in UV-Vis spectroscopy signifies the selective energy absorption by electrons at a particular wavelength. This absorption phenomenon indicates that electrons transition from their lowest energy level (ground state) to a higher energy level (excited state). These transitions provide insights into a band gap within the material, wherein electrons shift from a lower energy state to a higher energy state upon exposure to absorption wavelength. The material's corresponding absorption wavelength can ascertain the magnitude of this band gap.

The energy equation of quantum mechanics:

$$E_g = h * C / \lambda \quad (10)$$

Where, E_g is Bandgap (eV), h is Planck constant, 6.626×10^{-34} J s, C is velocity of light, 2.99×10^8 m/s, and λ (Wavelength) is Absorption peak

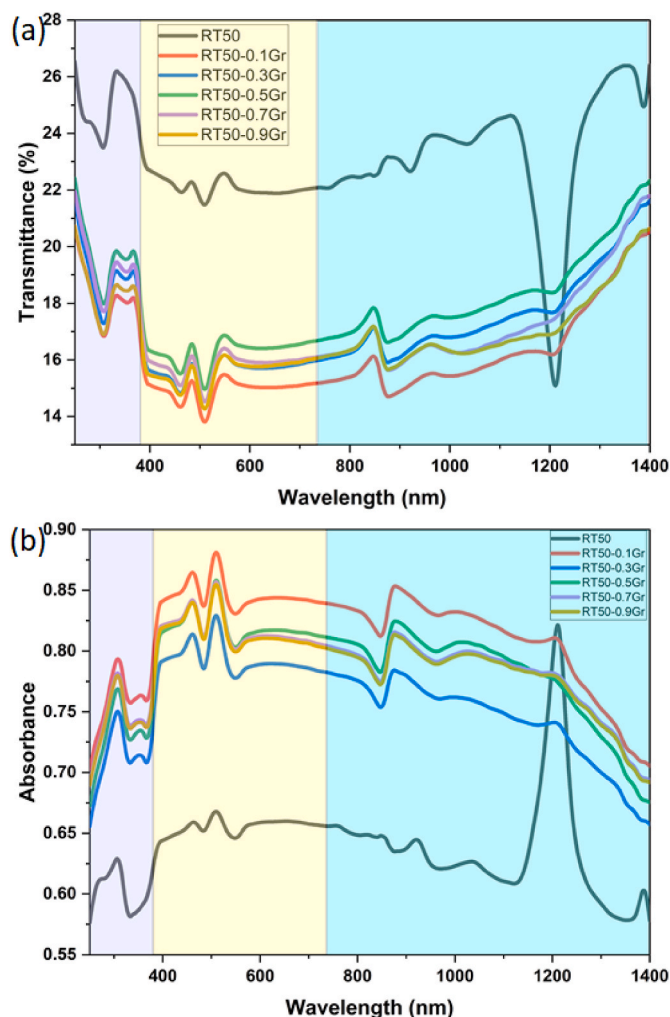


Fig. 7. Optical performance (a) Transmittance of PCM with various concentrations of Gr, (b) Absorbance of PCM with various concentrations of Gr.

(nm). Also, $1 \text{ eV} = 1.6810^{-19} \text{ J}$ [39].

Similarly, the optical band gap has been evaluated mathematically from Tauc equation stated as equation (11) [39].

$$(\alpha h\nu)^n = C(h\nu - E_g) \quad (11)$$

Where, E_g = average band gap of the material, C = material constant, α = absorption coefficient, $h\nu$ = photon energy and $n = \frac{1}{2}$ (for indirect band gap), 2 (for direct band gap)

The determination of the average band gap was achieved by identifying the intercept of the linear segment of $(\alpha h\nu)^2$ versus $h\nu$, with the direct band gap values plotted on the Y-axis as $(\alpha h\nu)^2$ and photon energy ($h\nu$) on the X-axis, as illustrated in Fig. 8.

The Tauc method plots the square of the absorption coefficient $(\alpha h\nu)^2$ against the photon energy ($h\nu$). Equation (11) extracted the band gap values for PCMe and the composite materials. Fig. 8(a)–(f) visually represents the methodologies employed to quantify the bandgap values and their corresponding wavelengths within PCM and nanocomposite materials. Notably, the bandgap of pure RT50 PCM measured to be 2.25 eV. On the other hand, the bandgap values for the Gr-incorporated PCM composites RT50-0.1Gr, RT50-0.3Gr, RT50-0.5Gr, RT50-0.7Gr, and RT50-0.9Gr were found to be 1.75 eV, 1.8 eV, 1.8 eV, 2.0 eV, and 1.9 eV, respectively. The introduction of Gr into PCM nanocomposites reduced band gap values by 22.22 %, 20.0 %, 20.0 %, 11.11 %, and 15.5 %, respectively, when compared to the band gap of the reference salt hydrate. Tauc's analysis further reinforces that Gr dispersion effectively

diminishes the band gap, with the maximum reduction amounting to 22.22 %. Incorporating Gr scaffolds in the nanocomposites led to a decreased band gap, consequently establishing a more photo-responsive region. This enhanced region holds significant potential for photon generation and utilization, rendering these materials more suitable for photon-centric applications.

3.5. Thermal stability

A PCM composite with exceptional thermal stability in practical applications is highly desirable, as it enables usage at elevated operating temperatures, particularly in latent heat storage. To access the thermal stability of the newly developed nanocomposite, TGA was employed to monitor the weight percentage loss with increasing temperature. Fig. 9 depicts the gradual reduction in the nanocomposite's weight at various graphene concentrations up to 210 °C, exhibiting no noticeable weight loss. Subsequently, beyond this temperature, a gradual decrease in weight was observed. The most substantial weight loss happened between 210 °C and 320 °C. In comparison, the pure RT50 demonstrated a final degradation temperature of 309 °C. The final degradation temperatures for the developed composites-RT50-0.1Gr, RT50-0.3Gr, RT50-0.5Gr, RT50-0.7Gr, and RT50-0.9Gr were determined as 305 °C, 311 °C, 320 °C, 311 °C, and 313 °C, respectively. The increasing thermal stability is thermal barrier formation due to addition of graphene nanoparticles and decrease in thermal stability is generation of free oxygen and oxygen vacancies in graphene results thermal excitation [40]. Indeed, the weight loss cure of sample RT50-0.1Gr exhibits a slightly steeper slope compared to the pure PCM. This occurrence may be attributed to the presence of micro impurities in the selected sample or generation of free oxygen and oxygen vacancies in graphene results thermal excitation.

Furthermore, a notable observation emerged: the decomposition temperature of the developed nanocomposite observed surpassed that of RT50 PCM across all samples. This enhancement in decomposition temperature is attributed to the successful dispersion of nanoparticles onto RT50 PCM, which in turn improves the material's thermal stability. This finding aligns with previous experimental results [41]. The TGA curves also indicate the effective incorporation of Gr nanoparticles into the PCM matrix. Particularly noteworthy is the consistent behaviour exhibited by the prepared nanocomposite PCM when subjected to temperatures ranging from 150 °C. This attribute presents promising advantages for applying the composite PCM in diverse systems, including photovoltaic thermal systems, desalination systems, battery thermal management, and electronic cooling.

3.6. Thermal conductivity

The thermal conductivity of PCM is a crucial parameter that substantially impacts the heat transfer rate, heat storage, and the release of thermal energy. In this study, the thermal conductivity of the developed nanocomposites was measured using the TEMPOS dual needle SH-3 instrument at room temperature. Fig. 10 presents the experimental and numerical results of RT50 PCM infused with varying concentrations of Gr nanoparticles. The base PCM demonstrated a measured thermal conductivity of 0.23 W/mK. Meanwhile, the thermal conductivity of the nanocomposites ranged from 0.307 to 0.403 W/mK as shown in Fig. 10 (a).

It is noteworthy to mention that the enhancement in thermal conductivity across the nanocomposites compared to the base PCM was notable: 33.57 %, 42.17 %, 61.3 %, 75.09 %, and 55.22 % for RT50-0.1Gr, RT50-0.3Gr, RT50-0.5Gr, RT50-0.17Gr, and RT50-0.9Gr, respectively. The highest enhancement, 75.09 %, was obtained with 0.7 wt% of Gr. This improvement is attributed to the higher aspect ratio or Gr, resulting in a greater surface area-to-volume ratio, which fosters an efficient thermal network and augments thermal conductivity. However, nanoparticle agglomeration and clustering occur at higher

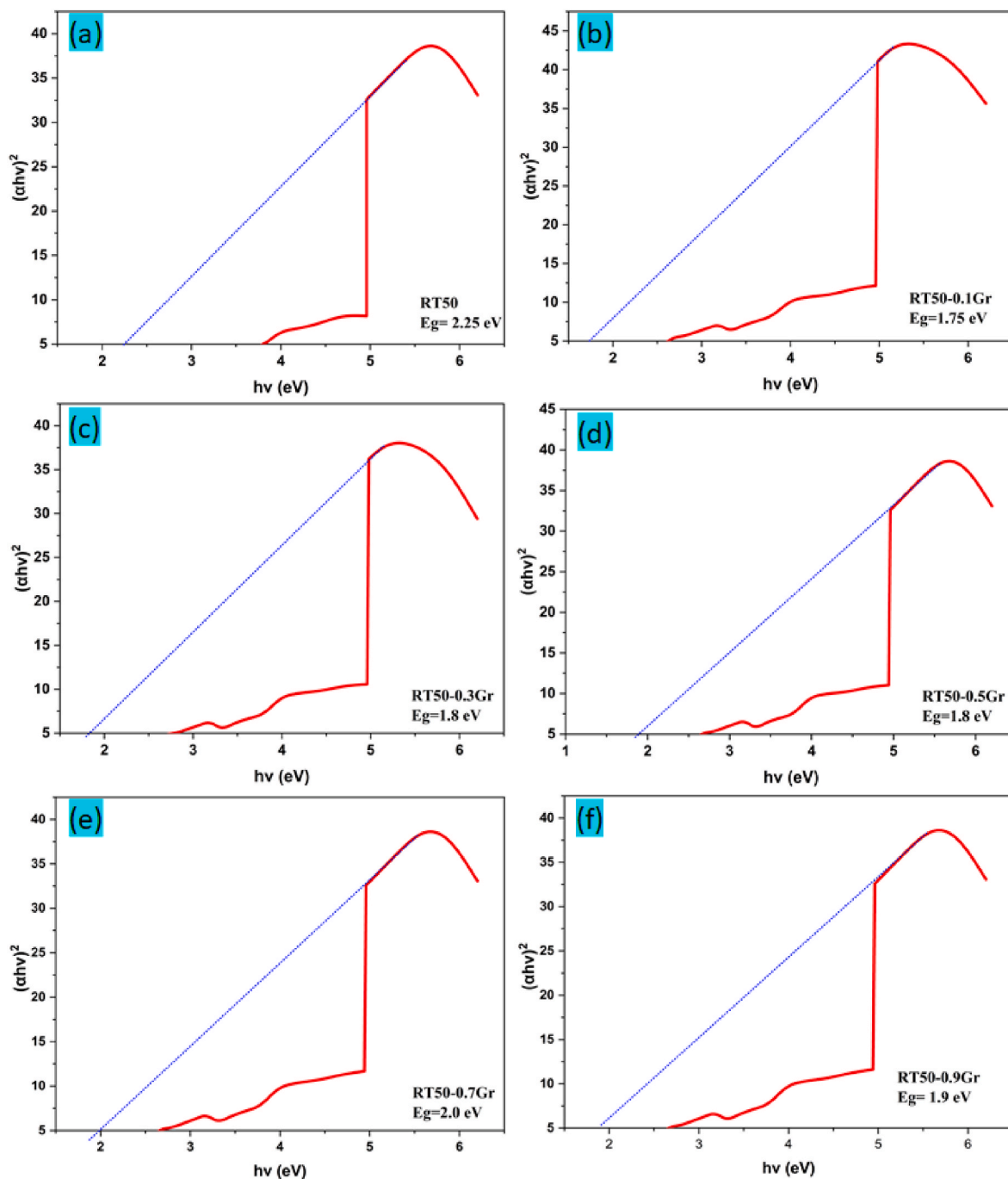


Fig. 8. Bandgap estimated using Tauc graph (a) Pure PCM, (b) RT50-0.1Gr nanocomposite, (c) RT50-0.3Gr nanocomposite, (d) RT50-0.5Gr nanocomposite, (e) RT50-0.7Gr nanocomposite, and (f) RT50-0.9Gr nanocomposite.

concentrations, leading to settling and a slight decline in thermal conductivity due to disrupted thermal networks.

The thermal conductivity of the samples is primarily influenced by two main mechanisms: phonon transfer and electron transfer. In the present context, phonon transfer plays a dominant role between PCM and Gr nanoparticles, while electron transfer has a minor effect. At lower concentrations, there might not be a well-established nanoparticle network. However, a well-organized thermal network forms as the concentration increases, elevating thermal conductivity. However, at higher concentrations, factors such as phonon-phonon scattering and phonon-impurity scattering come into play, shortening the phonon's path and decreasing thermal conductivity [40]. Moreover, an excess of nanoparticles introduced into the PCM leads to nanoparticle

accumulation, causing a decrease in thermal conductivity. From the formulated nanocomposite, the RT50 with 0.7 wt% Gr composition stands out as the most suitable nanocomposite for thermal energy storage applications.

The thermal conductivity of the PCM is inherently linked to the thermal conductance of the nanoparticles and the interfacial bonding between nanoparticles and PCM. Factors such as filler weight percentages, material density and nanoparticle shape significantly influence the thermal conductivity of the developed nanocomposite. Fig. 10(b) illustrates numerically calculated thermal conductivity values utilizing various models, including the parallel and series models, Hamilton, Maxwell, and Bruggeman models. The parallel model is particularly suitable for nanoparticle arrangement as it assumes a uniform dispersion

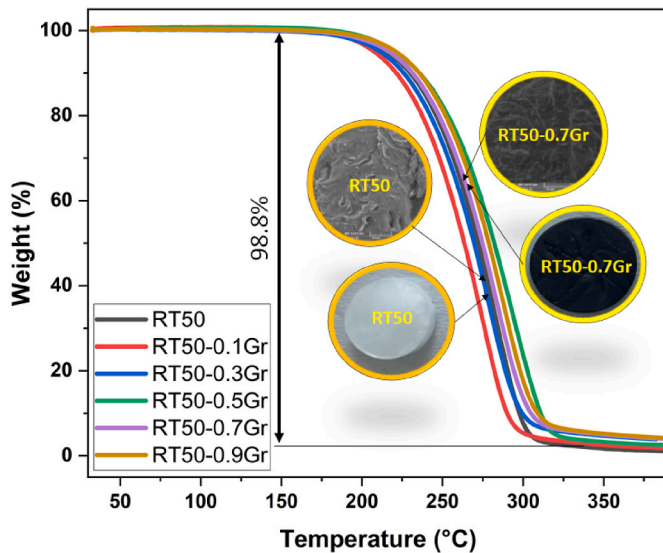


Fig. 9. Thermal degradation analysis of RT50 PCM with various percentages of Gr nanocomposites.

of nanoparticles within RT50 PCM, thereby creating an effective thermal network that facilitates heat transfer. Conversely, the series model, which accounts for nanoparticle accumulation, leads to lower thermal conductivity values. Both the parallel and series models derive from Fourier's heat conduction equation. The Maxwell model estimates thermal conductivity for cases where nanoparticles are randomly distributed in RT50 PCM. The Hamilton-modified model demonstrates enhanced thermal conductivity due to the uniform nanoparticle distribution. On the other hand, the Bruggeman model yields notably lower thermal conductivity values, predominantly applicable to two-phase polymeric composites. In summary, the parallel model is seen to be the most fitting approach for evaluating the thermal conductivity of Gr-dispersed PCM compared to the Bruggeman and Hamilton crossover models.

3.7. Melting enthalpy

DSC is a method employed to investigate PCM and nanocomposite samples' melting enthalpy and phase transition temperature. The vital measurable parameters derived are latent heat and phase transition temperature. Fig. 11(a) demonstrates the state of PCM/NeOPCM during melting and freezing behaviours. The graph in Fig. 11(b) presents the

melting enthalpy at the melting and freezing temperatures of both the base PCM and varying concentrations of Gr-enhanced PCM. Notably, the heating and cooling patterns for the base PCM and the nanocomposite exhibit dramatic similarities. Additionally, the graph illustrates two distinct phase change peaks during the melting process: a smaller initial peak indicating a solid-to-solid transition, followed by a broader peak corresponding to the solid-to-liquid transition.

Although the manufacturer specifies the phase change temperature of the PCM as 50 °C, in practice, the melting temperature of the samples tends to be slightly above 1 °C. This deviation could be attributed to experimental variables and the purity of the PCM obtained. However, it is worth highlighting that the phase transition temperatures observed during the heating process range from 52 °C to a freezing temperature of around 43.1 °C for the PCM. Similarly, the phase transition temperature of the nanocomposite during heating spans from 51.46 °C to 52.33 °C, while during freezing, it varies between 42.54 °C and 43.39 °C. In relation to latent heat, the measured melting enthalpy of the base PCM was 136 kJ/kg, which slightly contrasts with the manufacturer's indicated value of 146 kJ/kg. This discrepancy of 6 % could be attributed to experimental methodology and the PCM's purity. Nevertheless, this difference falls within an acceptable range. The latent heat values and the percentage enhancement in latent heat are presented in a bar chart, depicted in Fig. 11(c).

The obtained latent heat of the formulated nanocomposites (RT50-0.1Gr, RT50-0.3Gr, RT50-0.5Gr, RT50-0.7Gr, and RT50-0.9Gr) was 140.24, 144.32, 138.04, 137.93, and 136.8 kJ/kg, respectively. The addition of Gr nanoparticles to the PCM led to an increase in latent heat across all cases. The most significant improvement in latent heat was observed in the RT50-0.3Gr nanocomposite, with a 5.65 % increase. This marginal increment can be attributed to heterogeneous nucleation during the solidification process. Furthermore, the intermolecular interactions between Gr and RT50 PCM enhance the PCM's melting enthalpy during phase transitions [42]. Based on this analysis, it is evident that the prepared nanocomposites exhibit enhanced thermo-physical properties that are well-suited for efficient thermal energy storage applications.

3.8. Thermal reliability

The thermal cycling test was performed to verify the thermal reliability of the samples. The thermal cycler utilized for this study was designed with three separate compartments, allowing concurrent processing of three individual samples. The samples were placed within crucibles, each equipped with an electronically calibrated thermocouple. The cycler's maximum temperature was set at 70 °C, higher than

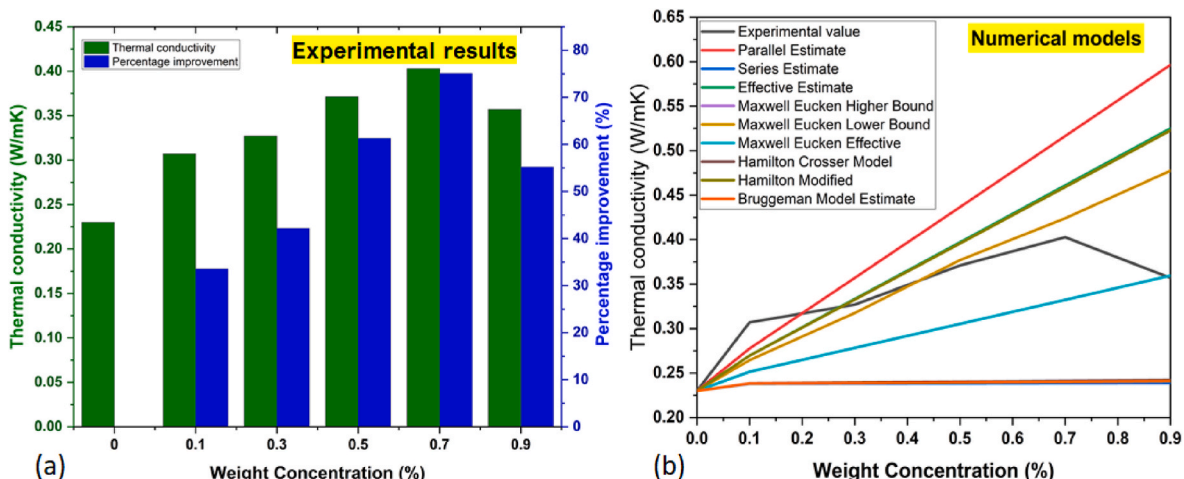


Fig. 10. Thermal conductivity analysis of RT50 with various percentages of Gr (a) experiment results, (b) Numerical results.

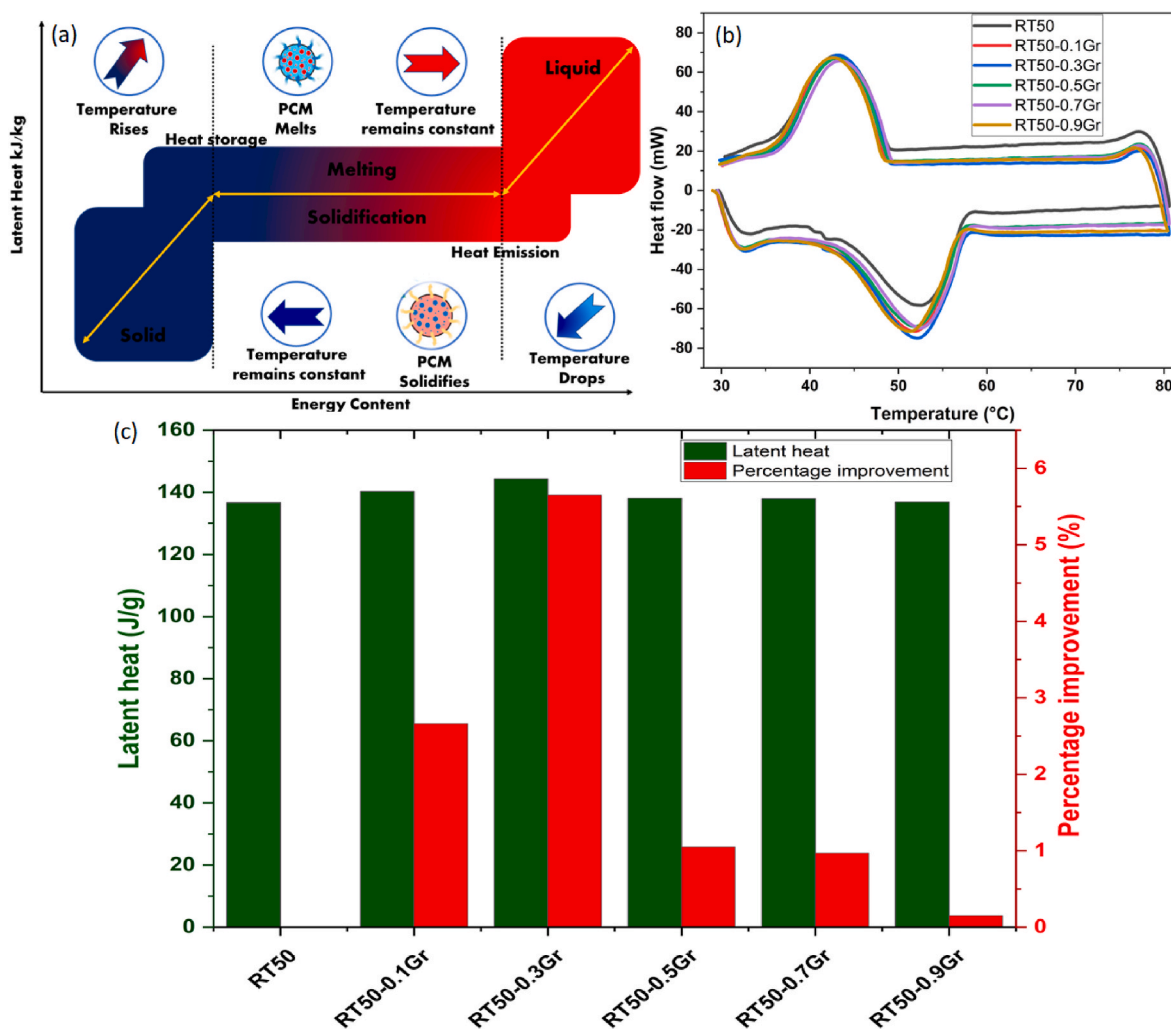


Fig. 11. (a) Energy storage mechanism of PCM during solid to liquid, (b) melting enthalpy and melting temperature graph, and (c) Latent heat value and percentage increment of RT50 with various concentrations of Gr.

the PCM's melting temperature. The crucibles were positioned beneath the sample holders to ensure optimal temperature distribution. The heater will be automatically deactivated upon reaching the maximum temperature, initiating the cooling cycle. Subsequently, the samples were cooled to 30 °C with the assistance of an electrically powered fan. The small sample quantity (around 2 g) facilitates rapid temperature changes within 2 min, followed by a 3-min cooling interval.

After the thermal cycling process, the samples were assessed for their chemical and thermal stability using FTIR and TGA. Fig. 12(a) confirms that the PCM and NeOPCM exhibit a stable dispersion, with no evidence of chemical reactions between Gr nanoparticles and PCM. Vibrational peaks at 2917 cm^{-1} , 2844 cm^{-1} , 1464 cm^{-1} , and 718 cm^{-1} persisted even after 3500 thermal cycles, resembling the peaks observed before the thermal cycling process. To offer a more precise insight, Fig. 12(c)–(f) presents digital images of the PCM and the nanocomposite before and after undergoing 500 thermal cycles. While slight colour changes were discernible after 500 cycles, they can be attributed to the increased number of heating and cooling cycles.

Fig. 12(b) shows the thermal decomposition temperatures of the base PCM and the RT50-0.7Gr nanocomposite both before and after thermal cycling. Before the thermal cycling, the final decomposition temperature of PCM and the RT50-0.7Gr composite were 309 °C and 311 °C, respectively. After 500 thermal cycles, the decomposition temperature of PCM and the RT50-0.7Gr composite increased to 315 °C and 311.94 °C, respectively. This observation suggests that the PCM and the

nanocomposite maintained or slightly increased their decomposition temperatures after thermal cycling. This demonstrates the sustained thermal stability of the developed nanocomposite PCM even after undergoing 500 cycles. The increase in decomposition temperature post-thermal cycling can be attributed to the repeated utilization of the PCM at elevated temperatures [43,44]. The higher decomposition temperature observed for the PCM indicates its potential for use in subsequent charging and discharging cycles. The results suggest that the NeOPCM composite remains remarkably stable after undergoing 500 cycles, rendering it a suitable candidate for medium-temperature solar thermal applications.

3.9. Numerical heat transfer simulation of PCM and nanocomposite

Based on the comprehensive characterization findings, it becomes apparent that the nanocomposite containing 0.7 wt% Gr (RT50-0.7Gr) exhibited optimal performance compared to other nanocomposites. Hence, the RT50-0.7Gr nanocomposite is chosen for numerical heat transfer analysis in this study. The analysis conducted in this study focuses on steady-state heat transfer, where the heat sink's temperature is maintained consistently across all studies.

Fig. 13(a)–(d) provide insight into the heat flow distribution within the 2D enclosure at various heat inputs: 55, 65, 75, and 85 °C. These diagrams illustrate how heat energy is distributed across the enclosure from the top surface to the bottom surface. For the analysis, the

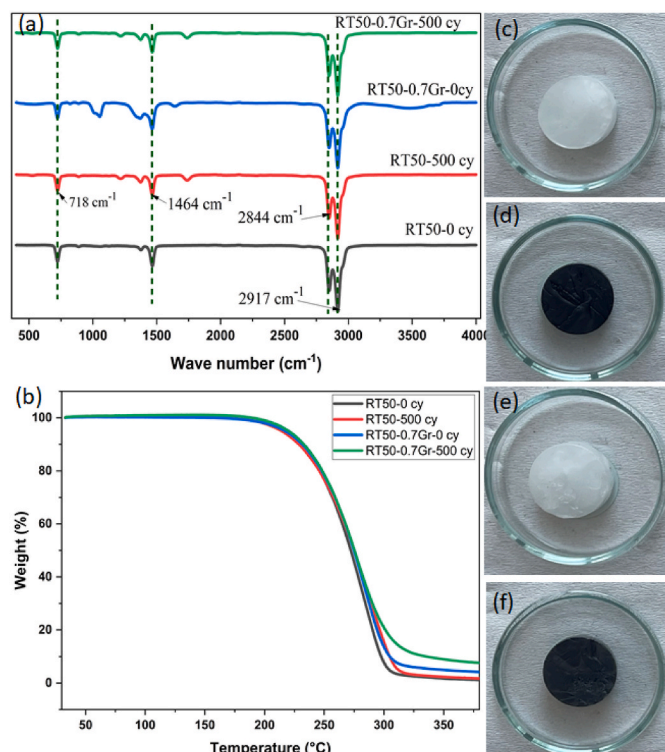


Fig. 12. Thermal reliability of PCM and RT50-0.7Gr nanocomposite after 500 cycles (a) Chemical stability, (b) Thermal stability, (c) digital image of pure RT50 PCM, (d) Pure Rt50 PCM after 500 thermal cycles, (e) RT50-0.7Gr composite and (f) RT50-0.7Gr after 500 thermal cycles.

circulation of liquid motion at the top surface of the 2D enclosure is disregarded due to phase transitions occurring in the PCM. As the heat supply from the heat source increases, there is a noticeable enhancement in the maximum heat transfer to the bottom surface. This enhancement can be attributed to the surplus energy supplied in these studies.

Fig. 14(a)-(d) focus on the heat transfer flow within the NeOPCM enclosure. These diagrams display the directional flow of heat from the top to the bottom region of the 2D enclosure. An interesting observation is that, with an increase in heat supply from the heat source, there is a significant improvement in the heat transfer rate. Additionally, it is evident that the heat transfer flow within the NeOPCM enclosure is more pronounced compared to that within the PCM enclosure. This increase in heat transfer rate can be attributed to two primary factors: (a) the higher thermal conductivity of NeOPCM, which facilitates a well-developed thermal network for faster heat flow, and (b) the increase in thermal energy magnitude, as indicated by the contour temperature values.

From the analysis provided, it becomes apparent that the depth of heat penetration and the rate of heat transfer exhibit variations for both PCM and NeOPCM. Moreover, the dispersion of Gr nanoparticles within the PCM leads to an increase in thermal conductivity, thus further enhancing the heat transfer rate. In light of these findings, it is established that achieving effective heat transfer requires the addition of the input heat source.

3.10. Application of Gr dispersed RT50 PCM nanocomposite PCM in photovoltaic thermal system

The energy performance analysis was conducted using an indoor PVT simulator under specific conditions, with an irradiance level of 800 W/m² and a flow rate of 0.0125 kg/s. These conditions were carefully selected to evaluate the integrated PVT system featuring NeOPCM accurately. This study section is dedicated to integrating the developed NeOPCM with the PVT system. The analysis encompassed various

aspects of energy performance, including panel temperature, electrical energy generation, thermal energy storage, and overall system efficiency. A comparative analysis was also conducted to provide a comprehensive assessment. This comparison involved evaluating the performance of the PVT-NeOPCM system concerning conventional PV systems, PVT systems without PCM integration, and PVT systems integrated with traditional PCMs. By conducting this comprehensive analysis, the researchers aimed to gain insights into the energy efficiency and potential advantages of integrating NeOPCM within PVT systems. This evaluation would help determine whether the NeOPCM-enhanced PVT system could outperform other configurations in terms of both electrical and thermal energy utilization.

3.10.1. Module temperature

Various cooling methods were applied to assess their impact on energy performance in the evaluation of the PV panel's temperature. The PV panel was positioned within the simulator, and electrical and thermal connections were established to the PV analyzer and data logger. The initial temperature of the setup was set and maintained at 25 °C. Subsequently, the PV panel was subjected to solar radiation at an intensity of 800 W/m², and the irradiance was measured by using a pyranometer.

Fig. 15 illustrates the bar chart depicting the panel temperature and the outlet temperature of the thermal fluid (water). The highest temperature recorded for the PV panel was 65.85 °C. Similarly, the PVT, PVT-PCM, and PVT-NeOPCM systems' measured surface temperatures were 50.95 °C, 47.75 °C, and 46.50 °C, respectively. Impressively, the integration of the NeOPCM-enhanced PVT system demonstrated a substantial reduction of 22.63 % compared to the standard PV system. This temperature decrease is attributed to the thermal fluid's ability to extract heat from the PV cell. The heat extraction efficiency hinges on factors like the fluid's thermal conductivity and flow rate.

Furthermore, when comparing with the PV system, the PVT-PCM and PVT-NeOPCM systems demonstrated percentage temperature reductions of 27.49 % and 29.38 %, respectively. The temperature decreases in the PCM and NeOPCM-integrated PVT systems arises from the cumulative cooling effects of the thermal fluid and the PCM/NeOPCM material. NeOPCM's higher thermal conductivity enables it to outperform PCM, leading to more efficient heat extraction capabilities and an advantage over both PVT and PVT-PCM systems. Additionally, the outlet temperature of the PCM/NeOPCM integrated PVT system slightly surpassed that of the PVT system. This occurrence can be attributed to the heat retained within the PCM/NeOPCM material, with only the residual heat being transferred to the thermal fluid. It was also evident that the water temperature in the PVT-PCM systems exceeded that in the PVT-NeOPCM system. This difference is attributed to NeOPCM's higher thermal conductivity, which enhances heat extraction and retention capacities. Hence, the NeOPCM-integrated PVT system could capture and conserve a larger portion of heat energy, enabling its utilization during periods of limited solar energy availability.

3.10.2. Electrical efficiency

The PV analyzer played a crucial role in measuring essential parameters such as the integrated PVT system's short-circuit current, open-circuit voltage, and power output. Fig. 16 shows the electrical power and electrical efficiency across four distinct configurations: PV, PVT, PVT-PCM, and PVT-NeOPCM. The electrical power output for the conventional PV system was quantified as 31.25 W, yielding an efficiency of 6.62 %. In contrast, the PVT, PVT-PCM, and PVT-NeOPCM systems demonstrated higher power outputs and efficiencies: 33.25 W, 42.15 W, 42.5 W, 7.04 %, 8.93 %, and 9.0 %, respectively. It is important to note that the standard PV system exhibited lower efficiency due to elevated temperatures adversely affecting the PV panel's performance. Introducing thermal collectors in the PVT system improved efficiency, elevating it from 6.62 % to 7.04 %. Moreover, integrating thermal collectors and NeOPCM resulted in a further enhancement, achieving an

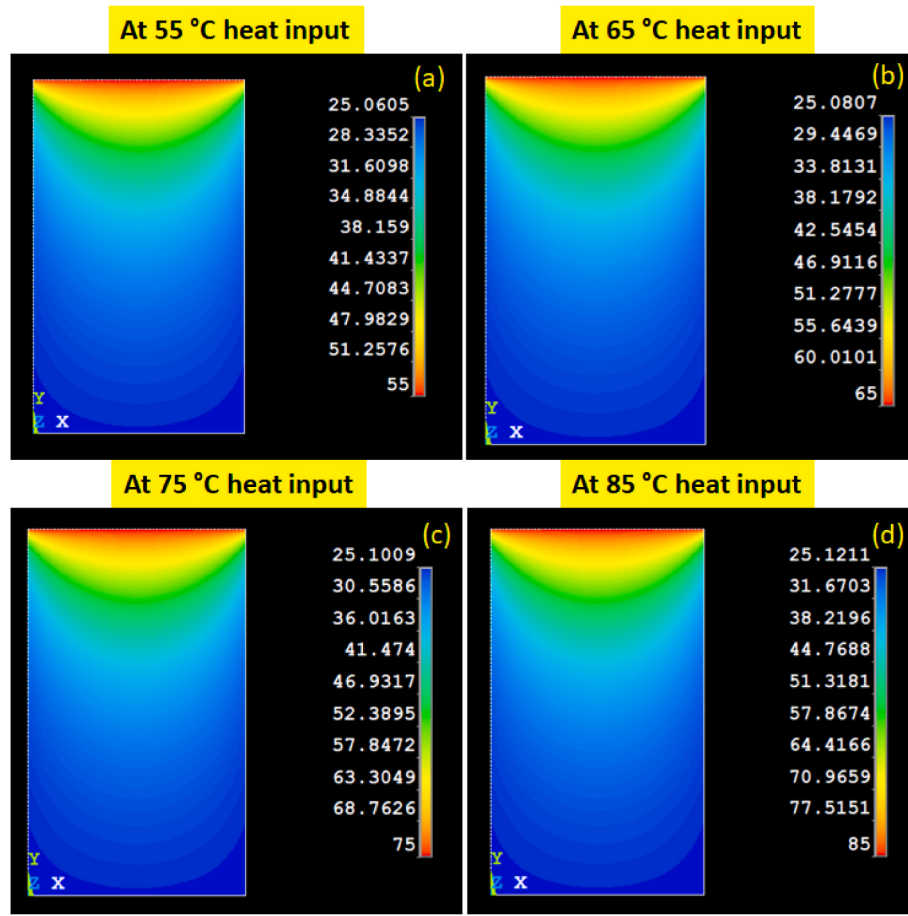


Fig. 13. Temperature contour plots of RT50 PCM at various heat input (a) RT50 @55 °C, (b) RT50 @65 °C, (c) RT50 @75 °C, and (d) RT50 @85 °C.

impressive efficiency of 9.0 %. This augmentation can be attributed to the synergistic cooling impact facilitated by the thermal fluid and the high thermal conductivity of the nano-enhanced PCM. Together, they effectively counteract the PV panel's temperature rise, thereby boosting its overall efficiency.

The interplay of heat transfer dynamics between the PV panel and NeOPCM hinges significantly on the thermal conductivity of the PCM. Notably, incorporating even a small quantity of nanoparticles can substantially increase thermal conductivity, subsequently enhancing the efficiency of heat dissipation and thermal management within the system.

3.10.3. Thermal efficiency

Evaluating heat gain and thermal energy involves considering both the cooling fluid's exit temperature and the accumulated thermal energy. Fig. 17 visually represents the heat gain and thermal efficiency across three distinct systems: PVT, PVT-PCM, and PVT-NeOPCM. The calculated heat gain values for the PVT, PVT-PCM, and PVT-NeOPCM systems were 9.96 W, 348.2 W, and 353.82 W, respectively. Notably, the PVT system integrated with NeOPCM exhibited a higher heat gain than the one integrated with PCM. This difference in heat gain can be attributed to the enhanced thermal conductivity of NeOPCM. The pivotal role of thermal conductivity is its impact on heat transfer efficiency and energy storage capacity. The elevated thermal conductivity of NeOPCM translates into improved energy storage potential within the system, offering a significant advantage over traditional PCM-integrated PVT systems.

In a conventional PVT system, the extracted heat energy is utilized immediately or lost, which signifies an efficiency drawback. In contrast, integrating PCM or NeOPCM into the PVT system allows for storing the

extracted energy within the phase change material. This stored energy can be effectively utilized during reduced solar energy availability, thereby minimizing energy wastage and enhancing overall system efficiency.

Furthermore, the PVT system integrated with NeOPCM achieved a remarkable maximum thermal efficiency improvement of 74.96 %, while the standard PVT system lagged at 69.9 %. Despite the slightly lower water outlet temperature in the PCM/NeOPCM-integrated PVT system, the utilization of stored heat within the PCM/NeOPCM material contributes to the pronounced enhancement in thermal efficiency. This improvement can be primarily attributed to the higher thermal conductivity exhibited by NeOPCM. Consequently, the results underscore the clear superiority of the PVT-NeOPCM system. These findings closely align with previous experimental investigations, corroborating the obtained outcomes' validity and significance.

3.10.4. Overall efficiency

Fig. 18 illustrates the overall efficiency of the PVT system, operating with a flow rate of 0.075 kg/s and subjected to solar radiation of 800 W/m². In this context, the PV system attained an electrical efficiency of 7.9 %. Remarkably, the PVT system integrated with NeOPCM achieved a peak efficiency of 9.1 %. Moreover, this NeOPCM-enhanced PVT system demonstrated an impressive thermal efficiency of 74 %. The overall efficiencies for various configurations are as follows: PV system: 7.9 %, PVT system: 8.1 %, PVT-PCM system: 78 %, and PVT-NeOPCM system: 81.9 %. Importantly, the PVT system integrated with NeOPCM showed the highest overall efficiency among these configurations. This achievement can be attributed to the meticulous design of the thermal collector within the PVT system, coupled with the successful integration of NeOPCM on the panel's rear side. The design of the collector involves

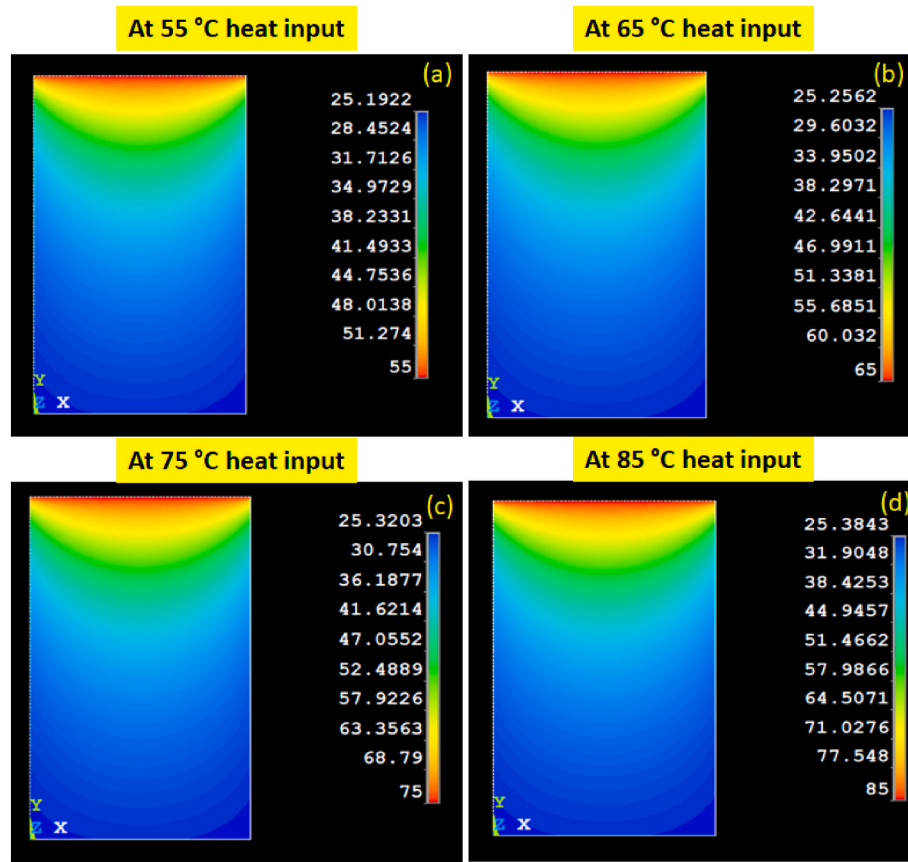


Fig. 14. Temperature contour plots of RT50-0.7Gr nanocomposite sample at various heat input (a) RT50-0.7Gr @55 °C, (b) RT50-0.7Gr @65 °C, (c) RT50-0.7Gr @75 °C, and (d) RT50-0.7Gr @85 °C.

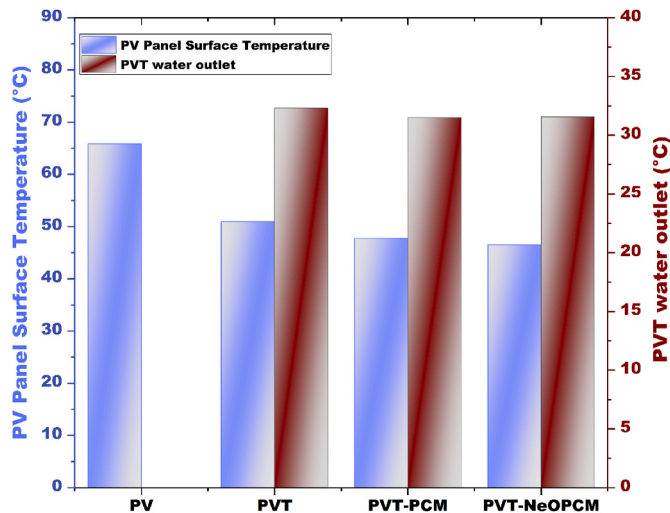


Fig. 15. Panel temperature and water outlet temperature of PV, PVT, PVT-PCM and PVT-NeOPCM system.

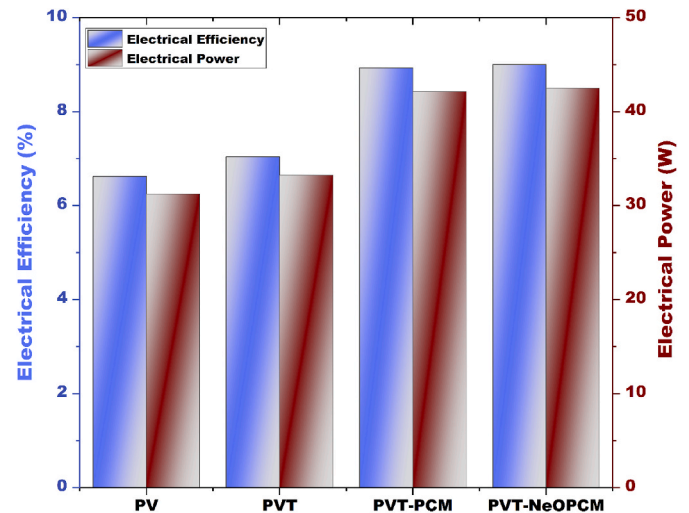


Fig. 16. Electrical efficiency of PV, PVT, PVT-PCM and PVT-NeOPCM system.

the strategic utilization of an elongated tube with smaller dimensions in terms of diameter and thickness. Integrating PCM/NeOPCM into PVT systems optimizes efficiency through the synergistic effects of well-engineered designs and inherent material properties [45]. Empirical observations underscore the significant potential of NeOPCM as a highly effective material for thermal management within photovoltaic systems. NeOPCM's exceptional capabilities in thermal energy storage contribute significantly to the elevation of both electrical and thermal efficiency

levels, ultimately enhancing the system's overall performance.

4. Conclusion and future research

The study comprehensively analyzed Gr nanopowder-dispersed RT50 PCM at various weight percentages, investigating its microstructure, optical properties, bandgap, stability, thermal properties, and numerical heat transfer analysis. In addition, the formulated nanocomposite was integrated into a PVT system and the energy

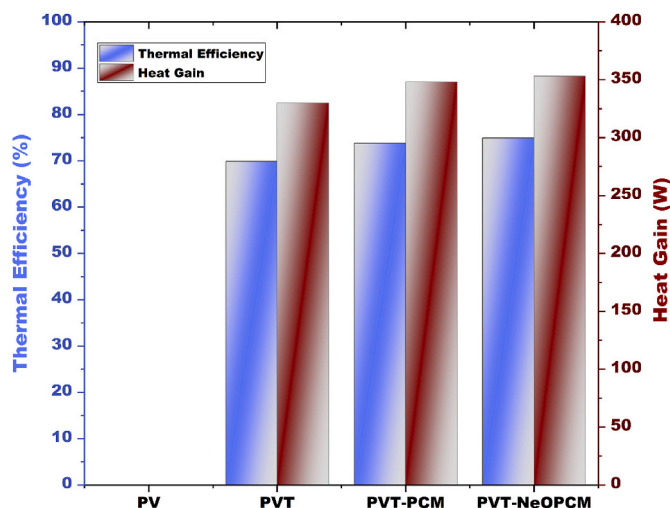


Fig. 17. Thermal efficiency of PV, PVT, PVT-PCM and PVT-NeOPCM system.

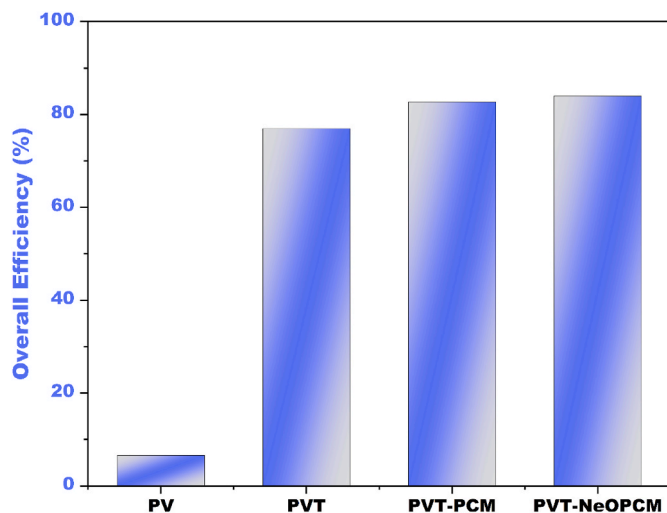


Fig. 18. Overall efficiency of PV, PVT, PVT-PCM and PVT-NeOPCM system.

performance were discussed. The key findings can be summarized as follows.

- The addition of Gr nanoparticles led to a substantial increase in thermal conductivity, with the highest enhancement achieved at 0.7 wt%. This improvement was attributed to improved thermal network formation and enhanced phonon interactions.
- The developed nanocomposites exhibited excellent thermal stability at temperatures up to 210 °C, with a slight enhancement observed upon the addition of nanoparticles. Importantly, no discernible chemical reactions occurred between PCM and the nanoparticles. The increasing thermal stability is due to thermal barrier formation due to addition of graphene nanoparticles and decrease in thermal stability is generation of free oxygen and oxygen vacancies in graphene results thermal excitation.
- The nanocomposite showed reduced light transmission and increased light absorbance, owing to decreased bandgap and improved light absorption capabilities due to the superior photo-thermal property. The latent heat of the nanocomposite exceeded that of the base PCM, attributed to heterogeneous nucleation during hardening step.

- The developed nanocomposites displayed thermal and chemical stability even after 500 thermal cycling operations, making them suitable material for energy storage applications.
- Numerical heat transfer analysis highlighted the significance of enhanced thermal conductivity towards efficient energy storage within a 2D enclosure.
- The integration of NeOPCM into a PVT system resulted decrease in the temperature of the PV panel, leading to improved electrical and thermal efficiencies.

This work offers significant insights into the utilization of nano enhanced phase change material in enhancing the efficiency of photo-voltaic thermal systems. Nevertheless, it is crucial to acknowledge that our study is impeded by a notable constraint that necessitate attention in subsequent investigations. The assumptions behind the simulation model employed to simulate the PCM/NeOPCM integrated PVT system may not entirely reflect the real-world behavior of the system in the field. For instance, the supposition that a photovoltaic surface is clear of dust may not hold true in reality, thereby affecting the system's effectiveness. Future studies could explore advanced control strategies using artificial intelligence (AI) and machine learning to optimize energy conversion and storage systems, making them more cost-effective. The research has implications for achieving Sustainable Development Goals related to number 7 and 13, which focus on ensuring access to affordable and clean energy and climate change.

CRedit authorship contribution statement

Reji Kumar Rajamony: Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – original draft. **Johnny Koh Siaw Paw:** Funding acquisition, Methodology, Project administration. **A.K. Pandey:** Methodology, Project administration, Software, Supervision. **Yaw Chong Tak:** Investigation, Software. **Jagadeesh Pasupuleti:** Conceptualization, Data curation, Writing – review & editing. **Sieh Kiong Tiong:** Methodology, Supervision, Writing – review & editing. **Talal Yusaf:** Formal analysis, Methodology, Writing – review & editing. **M. Samykano:** Project administration, Software, Visualization, Writing – review & editing. **A.G.N. Sofiah:** Investigation, Software, Validation, Writing – review & editing. **B. Kalidasan:** Data curation, Methodology, Writing – original draft. **Oday A. Ahmed:** Formal analysis, Methodology, Validation, Writing – review & editing. **K. Kadirgama:** Conceptualization, Data curation, Software, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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